STRATA: Conformal Prediction with Heterogeneous Message-Passing Calibration for Infrastructure Risk Assessment

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Abstract

We study conformal prediction for heterogeneous graphs where node difficulty depends on both graph topology and node type. STRATA combines a heterogeneous graph neural network with Conformal Heterogeneous Message Passing (CHMP), which rescales conformal scores using frozen training residuals aggregated over typed neighborhoods. This preserves split-conformal validity while adapting interval width to local graph difficulty. Across 20 synthetic seeds, all calibrators maintain near-nominal 90% coverage, while an ensemble variant yields the narrowest intervals (0.743 ± 0.052). On two public MATPOWER-based benchmarks, ACTIVSg200 and IEEE 118, STRATA attains average marginal coverage of 0.935 and 0.898, respectively. Relative to a Mondrian baseline on these real-data runs, CHMP matches average coverage, produces a small width reduction on ACTIVSg200, and shows no consistent calibration gain on IEEE 118. Additional ablations show the expected width-coverage tradeoff as α varies, and a group-wise audit yields a worst-group coverage gap of 0.035 in a representative run. These results position STRATA as a general methodology for uncertainty-aware prediction on heterogeneous graph-structured data while clarifying that CHMP's empirical gains are strongest in the synthetic benchmark.

Keywords: conformal prediction, graph neural networks, heterogeneous graphs, uncertainty quantification, calibration

1 Introduction

Critical infrastructure systems—electric power, water distribution, and telecommunications—form interdependent networks where failures cascade across utility boundaries (Rinaldi et al., 2001; Buldyrev et al., 2010). A transformer outage at a power substation can disable water treatment pumps, which in turn disrupts telecom cooling systems at co-located facilities (Ouyang, 2014). Predicting node-level risk scores in such coupled systems requires not only accurate point predictions but also reliable uncertainty quantification: operators need to know
emphhow wrong a risk estimate might be at each infrastructure node, and that uncertainty is inherently non-uniform across the heterogeneous graph.

Conformal prediction (Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos and Bates, 2023) offers distribution-free prediction intervals with finite-sample coverage guarantees, making it attractive for safety-critical infrastructure applications where parametric assumptions may be violated. However, standard split conformal prediction produces inter-

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vals of uniform width across all nodes, ignoring the topology-dependent difficulty structure that arises from infrastructure coupling.

Graph neural networks (GNNs) (Kipf and Welling, 2017; Gilmer et al., 2017; Hamilton et al., 2017) have emerged as powerful tools for node-level prediction on relational data, and recent work has extended conformal prediction to graph-structured settings (Zargarbashi et al., 2023; Huang et al., 2024; Clarkson, 2023). However, existing approaches either treat the graph as homogeneous or apply conformal calibration independently per type using Mondrian splits (Vovk, 2012), which sacrifices statistical power through smaller per-type calibration sets.

We address these limitations with **STRATA**, a framework that integrates three key innovations:

1. **Conformal Heterogeneous Message Passing (CHMP)**: A propagation-aware calibration scheme that normalizes nonconformity scores using frozen training-set residuals propagated through the infrastructure coupling topology. Rather than reducing aggregate interval width, CHMP *redistributes* width across nodes—giving wider intervals to nodes in high-difficulty neighborhoods and narrower intervals to low-difficulty nodes.

2. **A family of advanced calibrators** that learn the normalization function from data: a MetaCalibrator using heteroscedastic Gaussian NLL (Kendall and Gal, 2017), an AttentionCalibrator using learned neighbor weights, a LearnableLambdaCalibrator with per-type grid-search, and a heterogeneous CQR variant (Romano et al., 2019). An ensemble-based calibrator using epistemic variance (Lakshminarayanan et al., 2017) achieves the strongest width reduction (6.4% narrower than Mondrian CP).

3. **Spatial diagnostics** including Moran's I autocorrelation tests (Moran, 1950), Getis-Ord G_i^* hotspot detection (Getis and Ord, 1992), and conformal kriging surfaces (Cressie, 1993) that bridge infrastructure risk assessment with geostatistical analysis.

2 Related Work

2.1 Conformal Prediction

Conformal prediction, developed by Vovk et al. (2005), provides distribution-free prediction regions with finite-sample coverage guarantees under the exchangeability assumption. The inductive (split) conformal framework (Papadopoulos et al., 2002) partitions data into training, calibration, and test sets. Lei et al. (2018) established distribution-free inference guarantees for regression. Mondrian conformal prediction (Vovk, 2012) extends coverage guarantees to group-conditional settings.

Recent extensions have relaxed the exchangeability requirement. Barber et al. (2023) proved finite-sample coverage results beyond exchangeability, directly motivating our use of normalized scores on graph-structured data. Gibbs and Candès (2021) proposed adaptive conformal inference for streaming settings with distribution shift. Prior impossibility results for exact conditional coverage without strong distributional assumptions motivate STRATA's approximate type-conditional approach via Mondrian splits combined with propagation-aware normalization.

2.2 Conformal Prediction on Graphs

Zargarbashi et al. (2023) adapted conformal prediction sets for GNN classification, while Huang et al. (2024) developed CF-GNN for node-level uncertainty quantification. Clarkson (2023) provided distribution-free coverage guarantees for node classification under graph dependence. STRATA differs by (i) targeting regression, (ii) modeling heterogeneous multi-typed infrastructure graphs, and (iii) introducing CHMP calibration.

2.3 Graph Neural Networks for Heterogeneous Data

Heterogeneous GNNs extend message passing to multi-typed graphs: R-GCN (Schlichtkrull et al., 2018) introduces relation-specific weight matrices, HAN (Wang et al., 2019) applies hierarchical attention, and HGT (Hu et al., 2020) uses transformer-style attention. STRATA's message passing layer follows the R-GCN design with per-edge-type linear transforms, residual connections, and dropout.

2.4 Uncertainty Quantification

Deep ensembles (Lakshminarayanan et al., 2017) provide calibrated epistemic uncertainty. Kendall and Gal (2017) decomposed uncertainty into aleatoric and epistemic components via heteroscedastic loss—directly inspiring STRATA's MetaCalibrator. Conformalized quantile regression (CQR) (Romano et al., 2019) combines neural network quantile regression (Koenker and Bassett, 1978) with conformal calibration. STRATA extends CQR to heterogeneous graphs with propagation-aware normalization.

3 Methods

3.1 Problem Formulation

Consider a heterogeneous infrastructure graph $G = (V, E, \tau_V, \tau_E)$ where $V = V_{\text{power}} \cup V_{\text{water}} \cup V_{\text{telecom}}$ with type function $\tau_V : V \to \{\text{power}, \text{water}, \text{telecom}\}$. Each node v_i has features $\mathbf{x}_i \in \mathbb{R}^d$ and a scalar risk label $y_i \in [0, 1]$.

Given a trained GNN with point predictions $\hat{y}_i$, we seek prediction intervals $C_i = [\ell_i, u_i]$ such that:

$$\Pr(y_i \in C_i) \geq 1 - \alpha \tag{1}$$

for a user-specified miscoverage level α, ideally with the additional type-conditional guarantee:

$$\Pr(y_i \in C_i \mid \tau_V(i) = t) \geq 1 - \alpha, \quad \forall t \in \{\text{power}, \text{water}, \text{telecom}\} \tag{2}$$

3.2 Heterogeneous GNN Architecture

STRATA's GNN backbone extends the R-GCN framework (Schlichtkrull et al., 2018) with residual connections. Each `HeteroMessagePassingLayer` ℓ computes:

$$\mathbf{h}_i^{(\ell)} = \text{ReLU}\left(\mathbf{W}_{\text{self}}\mathbf{h}_i^{(\ell-1)} + \sum_{r \in \mathcal{R}} \frac{1}{|\mathcal{N}_r(i)|} \sum_{j \in \mathcal{N}_r(i)} \mathbf{W}_r \mathbf{h}_j^{(\ell-1)}\right) + \mathbf{h}_i^{(\ell-1)} \tag{3}$$

where $\mathcal{R}$ is the set of edge types and $\mathbf{W}_r$ are per-relation weight matrices. Per-type output heads project final representations to scalar risk predictions.

3.3 Conformal Heterogeneous Message Passing (CHMP)

CHMP introduces propagation-aware normalization:

Step 1. After training, compute residuals $r_j = |y_j - \hat{y}_j|$ for all training nodes $j \in \mathcal{D}_{\text{train}}$.

Step 2. Aggregate neighbor difficulty:

$$\bar{r}_{\mathcal{N}(i)} = \text{agg}_{j \in \mathcal{N}(i) \cap \mathcal{D}_{\text{train}}}(r_j) \tag{4}$$

Step 3. Compute normalization signal: $\sigma_i = 1 + \lambda \cdot \bar{r}_{\mathcal{N}(i)}$

Step 4. Calibration scores: $s_i = |y_i - \hat{y}_i|/\sigma_i$

Step 5. Prediction intervals:

$$C_i = [\hat{y}_i - \hat{q}_{\tau(i)} \cdot \sigma_i,\ \hat{y}_i + \hat{q}_{\tau(i)} \cdot \sigma_i] \tag{5}$$

Theorem 1 (CHMP Coverage Guarantee) *Let $\{(X_i, Y_i)\}_{i=1}^{n+1}$ be exchangeable random variables, let $\hat{f}$ be trained on $\mathcal{D}_{train}$ (disjoint from calibration and test sets), and let $\sigma_i = 1 + \lambda \cdot \bar{r}_{\mathcal{N}(i)}$ where $\bar{r}_{\mathcal{N}(i)}$ is computed exclusively from frozen training-set residuals. Define normalized scores $s_i = |Y_i - \hat{f}(X_i)|/\sigma_i$ for $i \in \mathcal{D}_{cal} \cup \{n+1\}$. Let $\hat{q}$ be the $\lceil (1-\alpha)(|\mathcal{D}_{cal}| + 1)\rceil / |\mathcal{D}_{cal}|$ quantile. Then:*

$$\Pr\Big(Y_{n+1} \in [\hat{f}(X_{n+1}) - \hat{q} \cdot \sigma_{n+1},\ \hat{f}(X_{n+1}) + \hat{q} \cdot \sigma_{n+1}]\Big) \geq 1 - \alpha \tag{6}$$

Proof [Proof sketch] Since σ_i is a deterministic function of training data, the normalized scores s_i inherit exchangeability from the original variables (Barber et al., 2023). The result follows from the standard split conformal guarantee (Papadopoulos et al., 2002; Lei et al., 2018). The Mondrian extension to per-type quantiles similarly preserves type-conditional coverage (Vovk, 2012). ■

Assumption 1 *(A1) Exchangeability of calibration and test data conditioned on the training set. (A2) The normalization signal $\sigma_i \geq 1$ is computed solely from training-set quantities. (A3) For Mondrian type-conditional coverage, exchangeability holds within each infrastructure type.*

3.4 Advanced Calibrator Variants

MetaCalibrator. A lightweight MLP predicts σ_i from node-level features including the point prediction, neighbor residual statistics, and node degree, trained with heteroscedastic Gaussian NLL.

AttentionCalibrator. Learns attention weights over neighbors for difficulty aggregation via a small MLP.

LearnableLambdaCalibrator. Optimizes λ per type through a 50/50 calibration-tune split with grid search.

CQR with Propagation. Extends conformalized quantile regression (Romano et al., 2019) to heterogeneous graphs with optional σ_i normalization. Quantile heads predict $q_{\alpha/2}(\mathbf{h}_i)$ and $q_{1-\alpha/2}(\mathbf{h}_i)$ using pinball loss.

3.5 Ensemble Uncertainty Decomposition

M independent GNNs with different random seeds provide: $\hat{y}_i = \frac{1}{M}\sum_m \hat{y}_i^{(m)}$, $\mathrm{Var}_i = \frac{1}{M}\sum_m (\hat{y}_i^{(m)} - \hat{y}_i)^2$. The EnsembleCalibrator uses $\sigma_i = 1 + \lambda\sqrt{\mathrm{Var}_i}$.

4 Experimental Setup

4.1 Synthetic Infrastructure Data

We generate heterogeneous infrastructure graphs with 200 power nodes (tree topology), 150 water nodes (grid/mesh), 100 telecom nodes (star-hub), 8-dimensional features, and cross-utility coupling probability 0.3.

4.2 Real Infrastructure Data: ACTIVSg200

We use the ACTIVSg200 200-bus Central Illinois power grid (Birchfield et al., 2017), constructing a three-layer heterogeneous graph with power (200 nodes), water ($\sim$108 nodes derived from demand centers), and telecom ($\sim$80 hub nodes).

4.3 Additional Public Benchmark: IEEE 118

To reduce the risk that conclusions depend on a single real-network topology, we also evaluate STRATA on the public IEEE 118-bus MATPOWER case. As with ACTIVSg200, we retain the real transmission-layer graph and derive water and telecom layers from demand centres, yielding a second public heterogeneous benchmark with a distinct topology and degree distribution.

4.4 Evaluation Protocol

All experiments use 20 random seeds with 60/20/20 train/cal/test splits. Metrics: marginal coverage, type-conditional coverage, mean interval width, ECE (Guo et al., 2017), paired Wilcoxon tests (Wilcoxon, 1945), and Friedman test (Friedman, 1937).

5 Results

5.1 Baseline Comparison

All methods achieve the nominal 90% coverage target. Coverage is statistically indistinguishable (Friedman $p = 0.93$), confirming CHMP preserves the conformal guarantee. CHMP's primary effect is width *redistribution* rather than aggregate reduction.

5.2 Advanced Calibrator Comparison

The ensemble-based calibrator achieves the narrowest intervals (0.743), a 6.4% reduction from Mondrian CP, while maintaining valid coverage (0.920). CQR with propagation pro-

Table 1: Baseline comparison (20 seeds, $\alpha = 0.10$).

Method	Coverage	Width	ECE
Mondrian CP	0.908 ± 0.032	0.794 ± 0.072	0.057 ± 0.022
CHMP (mean)	0.908 ± 0.030	0.793 ± 0.071	0.080 ± 0.020
CHMP (median)	0.908 ± 0.030	0.794 ± 0.071	0.078 ± 0.021
CHMP (med.+floor)	0.908 ± 0.032	0.794 ± 0.072	0.067 ± 0.023

Table 2: Full method comparison (20 seeds, $\alpha = 0.10$). Best width in **bold**.

Method	Coverage	Width	ECE
Mondrian CP	0.908 ± 0.032	0.794 ± 0.072	**0.057**
CHMP (mean)	0.908 ± 0.030	0.793 ± 0.071	0.080
CHMP (median)	0.908 ± 0.030	0.794 ± 0.071	0.078
CHMP (med.+floor)	0.908 ± 0.032	0.794 ± 0.072	0.067
MetaCalibrator	0.916 ± 0.027	0.792 ± 0.069	0.075
AttentionCalibr.	0.914 ± 0.027	0.790 ± 0.070	0.070
LearnableLambda	0.914 ± 0.026	0.788 ± 0.064	0.065
CQR + propagation	0.900 ± 0.034	0.838 ± 0.063	0.083
Ensemble (M=3)	0.920 ± 0.030	$\mathbf{0.743 \pm 0.052}$	0.073

duces the widest intervals (0.838), suggesting that quantile head training on frozen representations is insufficiently expressive for this heterogeneous graph structure.

5.3 Lambda Sensitivity

Table 3: Lambda sensitivity (20 seeds, $\alpha = 0.10$).

λ	Coverage	Width	ECE
0.0	0.916 ± 0.033	0.792 ± 0.078	0.056
0.1	0.916 ± 0.032	0.791 ± 0.077	0.069
0.3	0.916 ± 0.029	0.789 ± 0.075	0.067
0.5	0.915 ± 0.029	0.788 ± 0.073	0.069
1.0	0.912 ± 0.029	0.791 ± 0.071	0.071

Coverage remains stable across all λ values (within 0.4%). Width shows minimal sensitivity ($< 0.5\%$ variation), confirming robustness to hyperparameter misspecification.

5.4 Statistical Significance

Friedman test (coverage). Across all nine methods and 20 seeds: $\chi^2 = 3.05$, $p = 0.93$. We fail to reject the null hypothesis of equal coverage performance.

Friedman test (ECE). $\chi^2 = 34.3$, $p < 10^{-4}$, indicating highly significant differences in calibration quality. Pairwise Wilcoxon tests confirm Mondrian CP achieves significantly lower ECE than CHMP variants ($p < 0.001$).

Width. Only CQR ($p = 0.024$, wider) and Ensemble ($p = 0.019$, narrower) differ significantly from Mondrian CP.

5.5 Public Real-Data Benchmarks

Beyond the main synthetic benchmark, we evaluated Mondrian and CHMP on two public MATPOWER-derived heterogeneous graphs using three seeds each. On ACTIVSg200, both methods achieved average marginal coverage of 0.935; CHMP slightly reduced mean width relative to Mondrian (0.564 vs. 0.568) but had slightly higher ECE (0.073 vs. 0.068). On IEEE 118, both methods again achieved average marginal coverage of 0.898, while CHMP was marginally wider (0.805 vs. 0.803) and less well calibrated by ECE (0.105 vs. 0.069). The main conclusion from these public-network runs is therefore conservative: STRATA transfers across more than one real topology, but CHMP does not yet demonstrate a consistent out-of-sample advantage over the simpler Mondrian baseline on these datasets.

5.6 Alpha Ablation and Fairness Audit

The alpha ablation confirms the expected conformal tradeoff: average marginal coverage decreases from 0.974 at $\alpha = 0.05$ to 0.829 at $\alpha = 0.20$, while average width decreases from 1.166 to 0.604. This monotone trend is important because it demonstrates that STRATA's implementation respects the intended width–coverage control surface rather than producing unstable behavior under retuning. A representative group-wise audit at $\alpha = 0.10$ produced an overall coverage of 0.965 and a worst-group coverage gap of 0.035 across the three infrastructure types, suggesting mild but measurable group disparity that should be monitored in deployment.

6 Discussion

6.1 Width Redistribution vs. Aggregate Reduction

CHMP's primary effect is spatial redistribution of interval widths rather than aggregate width reduction. Across 20 synthetic seeds, CHMP and Mondrian CP produce nearly identical mean widths (~ 0.794), but the spatial distribution differs: CHMP assigns wider intervals to nodes in high-difficulty neighborhoods and narrower intervals elsewhere. The real-data comparisons make clear that this redistribution is not automatically beneficial in aggregate: the width effect is slightly favorable on ACTIVSg200 and slightly unfavorable on IEEE 118, with no corresponding improvement in marginal coverage. The strongest claim supported by the current evidence is therefore mechanistic rather than uniformly empirical: CHMP changes where interval budget is allocated.

6.2 Limitations

1. **No significant aggregate width improvement.** CHMP does not produce narrower intervals than Mondrian CP ($p > 0.50$). Only the ensemble method achieves significant width reduction (6.4%, $p = 0.019$) at 3× training cost.

2. **Lambda insensitivity.** Width varies $< 0.5\%$ across $\lambda \in [0, 1]$, suggesting the normalization signal is weak relative to the conformal quantile correction.

3. **CQR underperformance.** CQR produces the widest intervals (0.838) and lowest coverage (0.900).

4. **ECE trade-off.** CHMP variants exhibit higher ECE than Mondrian CP (0.067–0.080 vs. 0.057, $p < 0.01$).

5. **Synthetic data dominance.** The water and telecom layers in ACTIVSg200 and IEEE 118 are derived from power demand heuristics rather than independent utility datasets. This makes the public real-data evaluation useful for topology transfer, but not a full validation on independently measured multi-utility networks.

6. **Mixed real-data evidence.** On the two MATPOWER-based benchmarks, CHMP matches Mondrian coverage but does not produce a consistent width or ECE improvement. The method's empirical advantage over simpler baselines therefore remains benchmark-dependent.

7. **Scalability.** Current evaluation on graphs with ∼400 nodes.

8. **Static analysis.** STRATA provides static risk intervals; streaming infrastructure monitoring requires online conformal updates (Gibbs and Candès, 2021).

9. **Residual group disparity.** The fairness audit shows a worst-group coverage gap of roughly 0.035 in a representative run. This is not catastrophic, but it indicates that type-conditional monitoring should remain part of the evaluation and deployment workflow.

7 Conclusion

We introduced STRATA, a framework for distribution-free uncertainty quantification on heterogeneous infrastructure graphs. The core contribution—Conformal Heterogeneous Message Passing—provides locally adaptive prediction intervals by propagating frozen training-set difficulty signals through the infrastructure coupling topology, preserving finite-sample coverage guarantees. Experiments across 20 random seeds demonstrate that all calibrator variants maintain valid marginal and type-conditional coverage (Friedman $p = 0.93$).

CHMP's primary effect is spatial redistribution of interval widths rather than aggregate width reduction. Among all calibrators, the ensemble approach achieves the only statistically significant width improvement (6.4%, $p = 0.019$). STRATA establishes a principled framework for applying conformal prediction to heterogeneous graph-structured data with per-type coverage guarantees, spatial diagnostics, and a family of normalization strategies. The current evidence supports CHMP as a plausible mechanism for difficulty-aware width

redistribution, while also indicating that simpler baselines remain competitive on public real-data benchmarks.

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Software. Code is available at `https://github.com/matthew-lottly/strata` under the MIT License.

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